Juan Pablo Ruiz Rodriguez

Data Scientist

20.11.1992
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https://www.linkedin.com/in/juanpabloruizr/
github.com/jpcosec

Profile

- Team worker
- Smart worker
- Technical proficiency
- Client relationship experience

EDUCATION

Electrical Engineering

Computational Intelligence
- **Note:** Degree program comprises >300 ECTS credits. Officially validated by Anabin (ZAB) as fully equivalent to a German Master's/Diplom degree.

2011 - 2019

Universidad de Chile

PUBLICATIONS

Research

Morphing Attack Detection with Incremental Synthetic Images

NeurIPS 2025

Workshop Paper

Research

Intelligent Tutor for Tasks in the Domain of Linear Algebra

ICITED 2025

Universidad Playa Ancha

EXPERIENCE

ML Consultant

Event-driven document processing with LLMs and Apache Airflow

7.2025-9.2025

AraraDS, Chile

Research Associate

Intelligent Tutoring System (ITS)
- **System Architecture:** Designing an end-to-end, multimodal ITS for mathematics education to decouple affective sensing, domain knowledge (RAG), and pedagogical decision-making.
- **Affective Computing:** Developing a real-time computer vision module (`LAETEC-Vision`) utilizing PyTorch and CNNs trained to classify student cognitive states.
- **HCI & Environment:** Built `CartesianPlayground`, an interactive geometric manipulation environment to serve as the primary educational interface.
- **GenAI Integration:** Prototyping a hybrid pedagogical engine leveraging LLMs and RAG to provide

11.2024 - Present

UPLA, Chile

ABOUT ME

Born: born 20.11.1992
Interests: Machine Learning, Music, Cuisine

LANGUAGES

Spanish *C2** *Native*

English *C1**

*Using CEFR rating for language proficency.

German *Basic Knowledge*

Python
JavaScript
Java

TECHNOLOGIES

Docker · GIT · OpenCV
Gemini · Tensorflow · Pytorch
Hadoop · Spark · Neo4J
MongoDB · Kakfa

REFEREES

Juan Pablo Morales
CTO at Kwali
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Miguel Alejandro Rodriguez
Researcher at Universidad de Playa Ancha
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